Feasible regions

1 The set of a ordered pairs that satisfy each inequality in the system.

2 C

3

a No solutions. The regions are on opposite sides of the boundary line.

b Infinitely many solutions. The regions have the infinite set of points on the boundary line in common.

4

a $x < 1,\ y \leq 3$

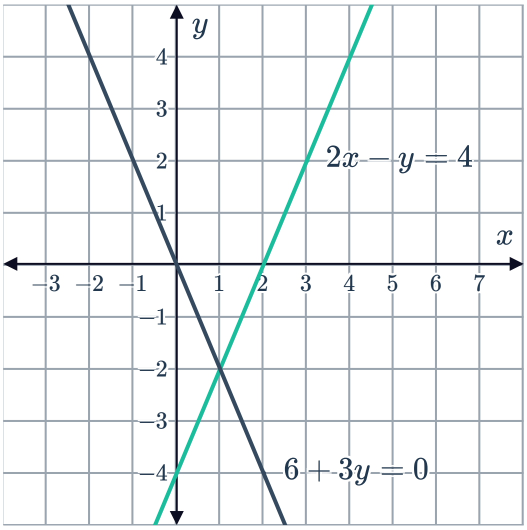
b $y \geq 2x - 3,\ y \leq 1$

c $x < 1,\ y < 1,\ y \geq -x - 5$

d $y + 3x < 0,\ y + 3x \leq 3$

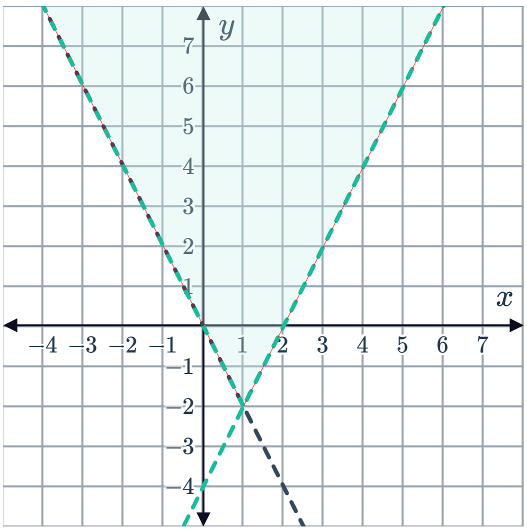
5

a

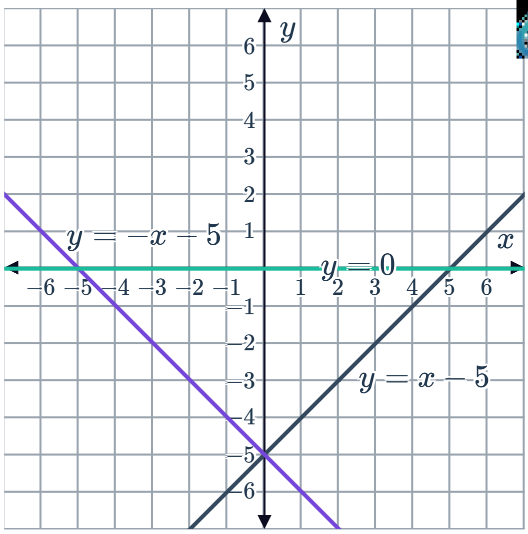


b Yes

c

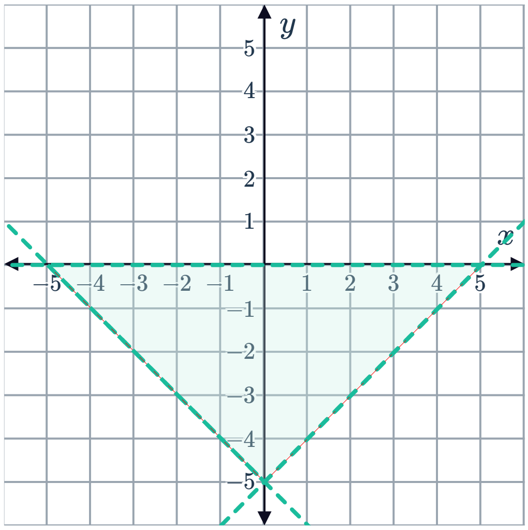


6 **a**



b Yes

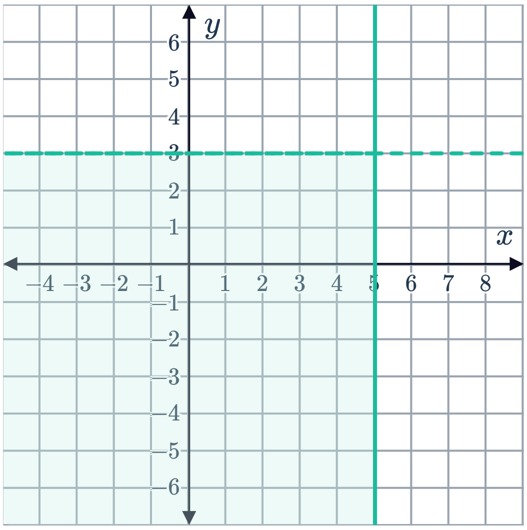
c



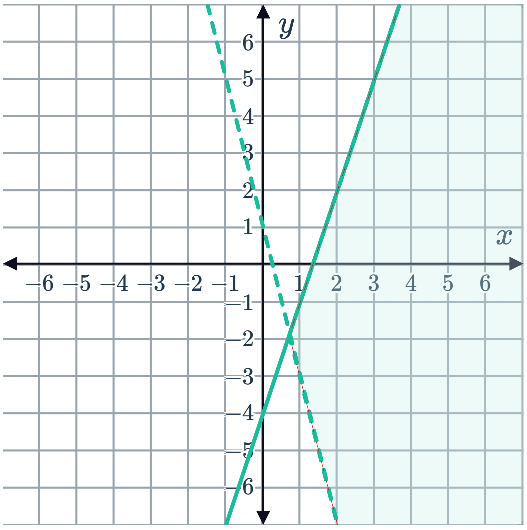
d 25 units^2

7

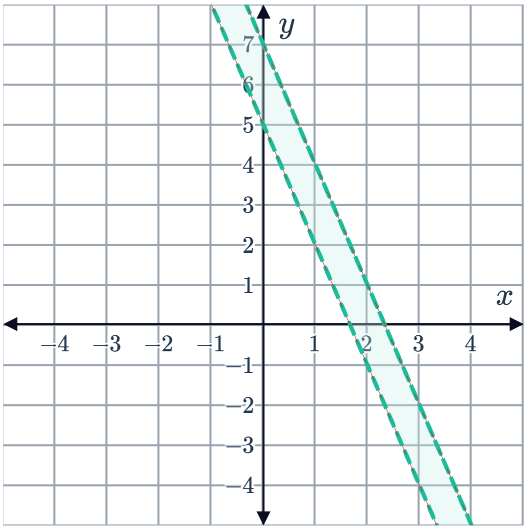
a



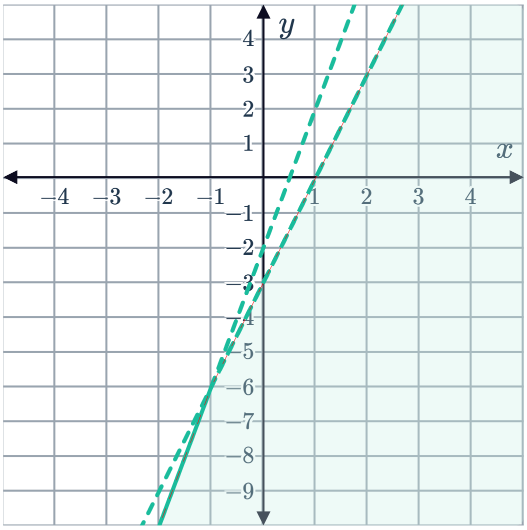
b



c

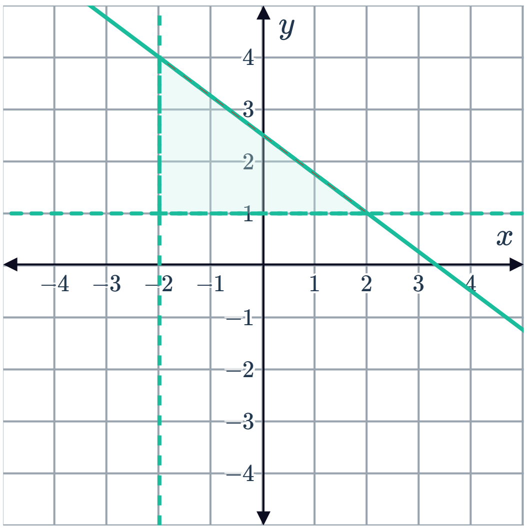


d



8

a



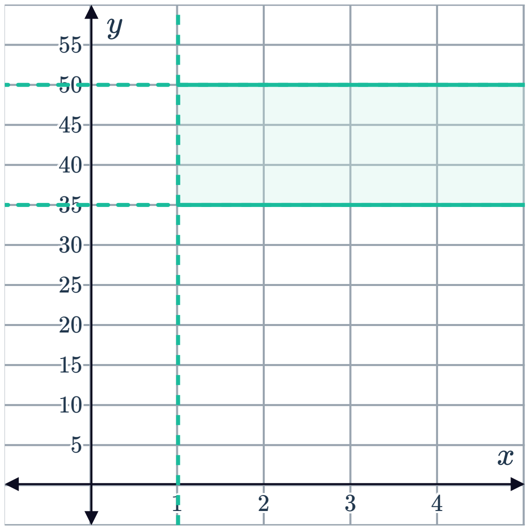
b 6 units^2

Applications

9

a $y \geq 35,\ y \leq 50$

b



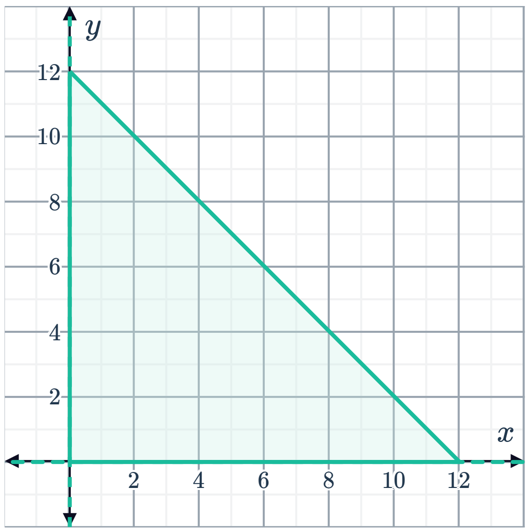
c Yes

d False

10

a $x \geq 0,\ 0 \leq y \leq 12 - x$

b



c

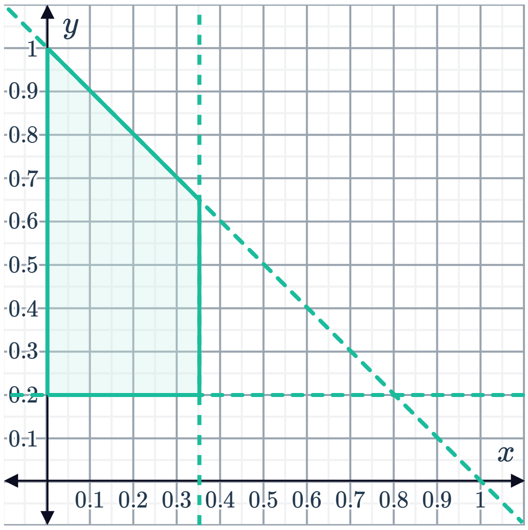
i Yes **ii** No **iii** No **iv** Yes

d $6 \text{ cm} \times 6 \text{ cm}$

11

a $0 \le x \le 0.35$, $y \ge 0.2$, $x + y \le 1$

b



c

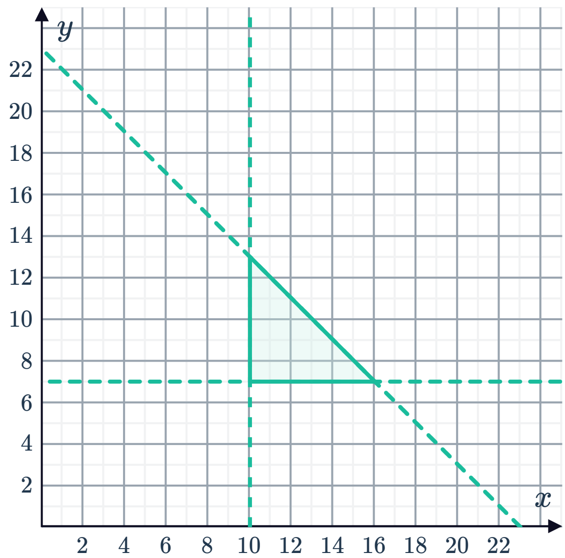
i Yes **ii** No **iii** Yes **iv** No

d Yes, because this would satisfy all of the inequalities and the point (0, 1) lies within the feasible region.

12

a $x \ge 10$, $y \ge 7$, $x + y \le 23$

b

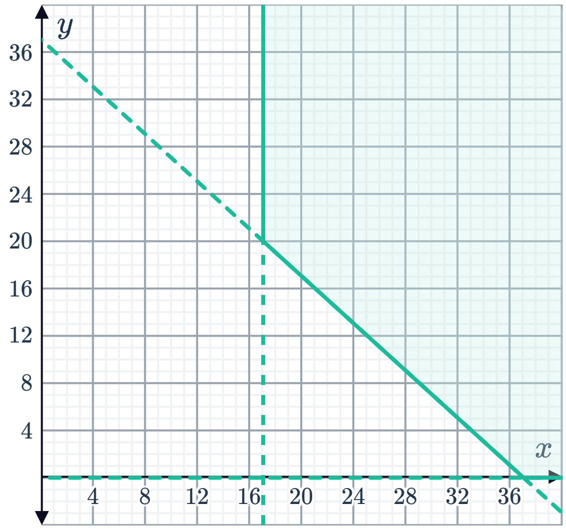


c 9

13

a $y \geq 0, x \geq 17, x+y \geq 37$

b



c

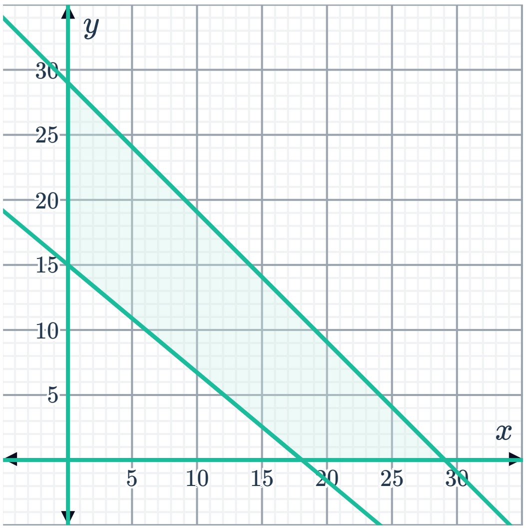
i No **ii** Yes **iii** No **iv** Yes

d 13

14

a $x \geq 0,\ y \geq 0,\ x+y \leq 29,\ 15x+18y \geq 270$

b



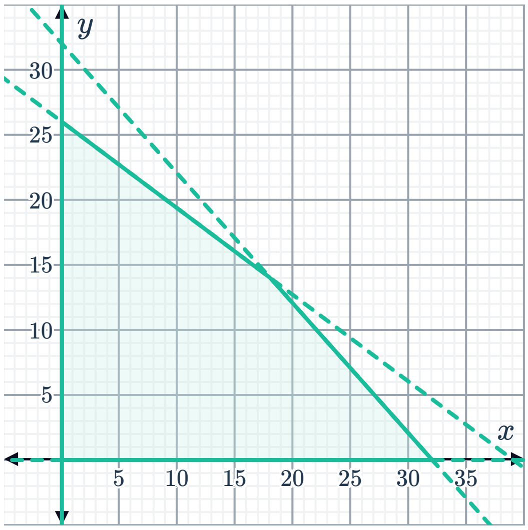
c 10 hours

d Yes

15

a $x \geq 0,\ y \geq 0, y \leq 26 - \frac{2}{3}x$ and $y \leq 32 - x$

b



c **A, B, C, D**